

Section A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1.				3d ↑↓↑↓↑↓↑↓ 4s ↑↓ 4p ↑↑↑	1			1		
2.				(+)6	1			1		
3.	(a)			$^{15}_7\text{N}$	1			1		
	(b)			$\frac{1}{32}$	1			1		
4.				Any of following - The second electron is being removed from a full shell that is nearer to the nucleus - The second electron is being removed from a positive ion - Greater effective nuclear charge on second electron - Less shielding since second electron in a new shell	1			1		
5.				HNO	1			1		
6.				63.1	1			1	1	
7.	(a)			$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$	1			1		
	(b)			$\frac{[\text{H}_2(\text{g})]}{[\text{I}_2(\text{g})]} = \frac{[\text{HI}(\text{g})]^2}{K_c} \quad (1)$ $= 15.00^2 / 1.20 \times 46.0 = 4.08 \text{ (mol dm}^{-3}\text{)} \quad (1)$ ecf possible from part (a)	1	1		2	1	
Section A total					9	1	0	10	2	0